

NELLORE, India

WMO: 432450

Lat: 14.450N

Lon: 79.983E

Elev: 66

StdP: 14.66

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	69.4	70.5	59.6	76.4	85.0	62.1	83.7	83.4	7.8	80.9	7.5	81.0	1.8	320	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.8	105.6	80.8	103.1	81.0	100.8	80.6	84.3	96.8	83.5	95.4	82.8	93.9	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	165.2	88.9	80.9	160.7	88.4	80.1	156.8	87.9	48.6	97.0	47.7	95.3	46.8	94.2	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	7.7	7.2	DB	66.1	110.4	2.9	1.7	64.0	111.7	62.3	112.7	60.7	113.6	58.6	114.9	
			WB	62.1	85.7	1.9	1.4	60.7	86.7	59.7	87.6	58.6	88.3	57.3	89.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	85.5	78.1	81.1	85.4	89.5	93.6	91.7	88.8	87.6	87.1	84.7	80.6	78.2
	DBStd	5.75	1.78	2.05	2.30	2.69	3.70	3.78	3.59	3.11	2.96	3.64	2.53	2.09
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12971	870	870	1096	1186	1351	1251	1203	1165	1114	1075	917	873
	CDD65	7496	405	450	631	736	886	801	738	700	664	610	467	408
	CDH74	93882	2896	4205	7457	10135	13392	12038	10537	9697	8870	7354	4359	2942
	CDH80	49673	743	1573	3522	5864	8955	7747	6149	5330	4645	3279	1252	614
(14) Wind	WSAvg	2.7	1.9	2.2	2.7	3.2	3.2	3.6	3.5	3.3	2.6	2.0	2.1	2.2
(15) Precipitation	PrecAvg	41.2	0.3	0.1	0.2	0.5	0.9	1.8	3.4	3.5	4.9	10.9	12.0	3.1
	PrecMax	68.6	1.5	0.5	3.4	3.9	4.9	5.6	6.9	12.2	14.2	29.8	30.4	13.0
	PrecMin	19.2	0.0	0.0	0.0	0.0	0.0	0.3	1.1	0.2	0.3	2.7	2.5	0.0
	PrecStd	12.5	0.4	0.2	0.7	0.9	1.2	1.3	1.5	3.1	3.2	7.8	8.0	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	94.2	99.9	105.5	109.6	106.7	101.7	99.7	99.6	98.0	91.9	88.0
		MCWB	73.2	74.7	77.8	80.4	80.4	81.4	79.8	79.2	78.7	77.2	76.7	74.7
	2%	DB	86.4	91.2	96.6	101.6	106.9	103.7	100.0	97.9	97.3	95.4	89.6	86.1
		MCWB	73.3	74.9	78.3	81.3	80.4	81.0	79.6	79.0	78.9	77.6	76.5	74.6
	5%	DB	85.3	89.3	94.5	99.3	104.7	101.5	98.2	96.4	95.6	93.5	87.7	84.9
		MCWB	73.1	74.7	78.1	80.9	80.9	80.6	79.4	78.9	78.9	78.0	76.5	74.1
	10%	DB	84.1	87.8	92.6	97.2	102.2	99.4	96.2	94.7	93.9	91.5	86.1	83.6
		MCWB	72.8	74.4	77.4	80.6	81.2	80.1	79.2	78.8	78.9	78.2	76.4	73.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	79.9	82.5	84.6	85.3	84.9	83.5	82.9	83.0	82.3	81.0	78.8
		MCDB	83.4	88.6	94.2	96.7	97.4	96.7	93.5	91.4	91.9	89.8	85.9	83.2
	2%	WB	76.3	78.3	81.4	83.5	84.5	83.7	82.3	81.8	81.8	81.4	79.8	77.4
		MCDB	82.3	86.1	91.6	95.0	97.2	97.0	92.8	90.2	90.4	88.5	84.8	81.9
	5%	WB	75.4	77.3	80.4	82.6	83.7	82.7	81.4	81.0	81.0	80.7	79.0	76.7
		MCDB	81.4	84.6	89.6	93.5	96.2	94.9	91.8	89.6	89.4	87.7	84.0	81.2
	10%	WB	74.4	76.4	79.5	81.9	83.0	81.9	80.7	80.4	80.5	80.0	78.4	75.9
		MCDB	80.6	83.6	87.9	92.1	94.7	93.5	90.9	89.2	88.7	86.8	83.1	80.4
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	15.5	17.2	17.6	17.8	14.9	13.1	12.9	13.0	11.8	10.5	11.3
		MCDBR	14.7	17.3	19.2	19.8	20.5	17.5	15.5	15.3	15.6	15.2	13.4	13.1
		MCWBR	4.8	5.1	5.5	5.2	5.8	5.5	5.0	4.6	4.8	4.5	4.3	4.4
	5% WB	MCDBR	12.3	14.9	17.4	18.3	18.6	16.2	14.6	14.0	13.9	12.5	9.9	9.8
		MCWBR	4.9	5.2	5.4	5.3	5.7	5.9	5.4	5.1	5.1	4.6	4.5	4.7
(40) Clear-Sky Solar Irradiance	taub		0.480	0.473	0.527	0.548	0.616	0.553	0.538	0.526	0.520	0.544	0.524	0.538
	taud		2.093	2.129	1.998	1.986	1.836	2.020	2.071	2.114	2.129	2.048	2.050	1.964
	Ebn at Noon		258	266	255	248	228	240	245	251	253	244	243	236
	Edh at Noon		50	50	58	59	67	56	53	51	50	54	52	56
(44) All-Sky Solar Radiation	RadAvg		1541	1842	2014	2094	1945	1651	1525	1545	1587	1454	1348	1331
	RadStd		57	63	62	64	68	122	144	77	102	124	119	108

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.41	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page